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LOS Assignment LAB-4

Part A – Basic grep and Filtering

Display all lines containing the word failed.

Display all lines containing failed, ignoring case.

Display all lines that contain either Failed or Denied.

Display all lines that do not contain the word INFO.

Display all lines containing the exact word root.

Display all occurrences of the word attack along with their line numbers.

Count the number of lines containing ERROR.

Display only the filenames of files whose contents contain the word malware, assuming multiple log files are available.

Display the first five lines containing Suspicious.

Display the last five failed-login entries.

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/ } grep "failed" security_log.txt

anonymous@anonymous-supreme in ~/Desktop as 
[*] / } grep -i "failed" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222

anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -E "Failed|Denied" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222

anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -v "INFO" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001

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anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -w "root" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:27:24] Suspicious crontab created for user root
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222

anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -nw "attack" security_log.txt

anonymous@anonymous-supreme in ~/Desktop as 
[*] / } grep -nw "attack" security_log.txt

anonymous@anonymous-supreme in ~/Desktop as 
[*] / } grep -c "ERROR" security_log.txt
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anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -l "malware" security_log.txt
security_log.txt

anonymous@anonymous-supreme in ~/Desktop as 
/ } grep "Suspicious" security_log.txt | head -5
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com/invoice_2025
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious attachment

anonymous@anonymous-supreme in ~/Desktop as 
/ } grep -i "failed" security_log.txt | tail -5
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222

anonymous@anonymous-supreme in ~/Desktop as 
/ }

```

Part B – Regular Expression Extraction

No.	Question	Command
1	Extract all IPv4 addresses from the log.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt
2	Extract all IP addresses belonging to the 10.x.x.x private network.	grep -Eo '\b10(\.[0-9]{1,3}){3}\b' security_log.txt
3	Extract all addresses belonging to the 192.168.x.x network.	grep -Eo '\b192\.168(\.[0-9]{1,3}){2}\b' security_log.txt
4	Extract all email addresses.	grep -Eo '[[[:alnum:]]._%+-]+@[[:alnum:]].-]+[[:alpha:]]+' security_log.txt
5	Extract only email addresses whose domain ends in .com.	grep -Eo '[[[:alnum:]]._%+-]+@[[:alnum:]].-]+\com\b' security_log.txt
6	Extract email addresses whose username contains at least one digit.	grep -Eo '[[[:alnum:]]._%+-]*[0-9][[:alnum:]]._%+-]*@[[:alnum:]].-]+[[:alpha:]]+' security_log.txt
7	Extract all URLs beginning with http:// or https://.	grep -Eo 'https?://[^\[:space:]]+' security_log.txt
8	Extract only HTTPS URLs.	grep -Eo 'https://[^\[:space:]]+' security_log.txt
9	Extract URLs ending with .php.	grep -Eo 'https?://[^\[:space:]]+\php\b' security_log.txt
10	Extract all CVE identifiers in the form CVE-YYYY-NNNN.	grep -Eo 'CVE-[0-9]{4}-[0-9]{4}' security_log.txt

11	Extract only CVEs reported for the year 2025.	grep -Eo 'CVE-2025-[0-9]{4}' security_log.txt
12	Extract all 32-character hexadecimal MD5 hashes.	grep -Eo '\b[0-9a-fA-F]{32}\b' security_log.txt
13	Extract all 40-character hexadecimal SHA-1 hashes.	grep -Eo '\b[0-9a-fA-F]{40}\b' security_log.txt
14	Extract all 64-character hexadecimal SHA-256 hashes.	grep -Eo '\b[0-9a-fA-F]{64}\b' security_log.txt
15	Extract all timestamps in the format YYYY-MM-DD HH:MM:SS.	grep -Eo '[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}' security_log.txt

Part C – Anchors and Regex Patterns

No.	Question	Command
1	Display all lines beginning with a timestamp.	grep -E '^'[0-9]{4}-[0-9]{2}-[0-9]{2}' security_log.txt
2	Display all lines beginning with the word Suspicious.	grep '^Suspicious' security_log.txt
3	Display lines ending with a numeric value.	grep -E '[0-9]+\$' security_log.txt
4	Display lines beginning with ERROR or WARNING.	grep -E '^(ERROR WARNING)' security_log.txt
5	Display lines whose first character is a digit.	grep -E '^'[0-9]' security_log.txt
6	Find lines ending in .exe.	grep -E '\.exe\$' security_log.txt
7	Find lines containing filenames ending in .exe, .sh, or .pdf.	grep -E '\.(exe sh pdf)\b' security_log.txt
8	Identify lines containing the exact word root, but not words such as rooted.	grep -w 'root' security_log.txt
9	Display blank lines in the file.	grep '^\$' security_log.txt
10	Display all nonblank lines.	grep -v '^\$' security_log.txt

Part D – Linux Filters and Pipelines

No.	Question	Command
1	Extract all IPv4 addresses, sort them, and remove duplicates.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort -u
2	Count the number of unique IPv4 addresses present in the log.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort -u wc -l
3	Extract all IP addresses and display them in sorted order.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort
4	Determine which IP addresses appear more than once.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort uniq -d
5	Determine the IP address that occurs most frequently.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort uniq -c sort -nr head -1
6	Extract all usernames from authentication entries and display unique usernames.	grep -i "authentication" security_log.txt grep -Eo 'user(name)?[=:]+[[[:alnum:]]_.-]+' sed -E 's/.*[=:]+//' sort -u
7	Count the number of authentication attempts associated with each username.	grep -i "authentication" security_log.txt grep -Eo 'user(name)?[=:]+[[[:alnum:]]_.-]+' sed -E 's/.*[=:]+//' sort uniq -c sort -nr
8	Extract all HTTP status codes and determine how many times each status code occurs.	grep -Eo '\b[1-5][0-9]{2}\b' security_log.txt sort uniq -c sort -nr
9	Display the five most frequently occurring IP addresses.	grep -Eo '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt sort uniq -c sort -nr head -5
10	Display the three most frequently occurring usernames.	grep -i "authentication" security_log.txt grep -Eo 'user(name)?[=:]+[[[:alnum:]]_.-]+' sed -E 's/.*[=:]+//' sort uniq -c sort -nr head -3
11	Extract all destination ports and display them in numerical order.	grep -Eo 'dst[_-]?port[=:]+[0-9]+' security_log.txt grep -Eo '[0-9]+' sort -n
12	Remove duplicate port numbers.	grep -Eo 'dst[_-]?port[=:]+[0-9]+' security_log.txt grep -Eo '[0-9]+' sort -nu
13	Count the number of unique destination ports.	grep -Eo 'dst[_-]?port[=:]+[0-9]+' security_log.txt grep -Eo '[0-9]+' sort -nu wc -l
14	Extract all file extensions appearing in the log and determine their frequency.	grep -Eo '\.[A-Za-z0-9]{1,5}\b' security_log.txt sed 's/^\./' sort uniq -c sort -nr
15	Find the five most frequently occurring domains.	grep -Eo '([[:alnum:]]-+\.)+[[:alpha:]]{2,}' security_log.txt sort uniq -c sort -nr head -5

No.	Task	Command
1	Both an IP address and Failed	grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt grep -i 'Failed'
2	Both root and Failed	grep -i 'root' security_log.txt grep -i 'Failed'
3	Either download or upload	grep -Ei 'download upload' security_log.txt
4	Either .exe or .sh	grep -Ei '\.exe \.sh' security_log.txt
5	Both a CVE identifier and an IP address	grep -E 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}'
6	Both a filename and a hash value	grep -E '[:alnum:]_-\+\.[:alnum:]]+' security_log.txt grep -Ei '\b[a-f0-9]{32,64}\b'
7	IP address followed by a port	grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}:[0-9]{1,5}' security_log.txt
8	HTTP request with URL and status code 404	grep -E 'https?:/[^\]+' security_log.txt grep -w '404'
9	Authentication failures excluding 127.0.0.1	grep -Ei 'authentication.*fail failed.*authentication' security_log.txt grep -v '127.0.0.1'
10	Attack-related entries excluding INFO	grep -Ei 'attack exploit malware intrusion brute.?force' security_log.txt grep -vi 'INFO'

Part F – Negative Matching

No.	Task	Command
1	Extract all IPv4 addresses except those beginning with 127.	grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt grep -v '^127\.'
2	Display all log entries except successful login events	grep -Eiv 'successful login' security_log.txt
3	Display entries that do not contain INFO, DEBUG, or SUCCESS	grep -Eiv 'INFO DEBUG SUCCESS' security_log.txt
4	Find HTTP requests whose status code is not 200	grep -E 'HTTP.*[^\0-9]([0-9]{3})[^\0-9]' security_log.txt grep -Ev '[:,space:]]200[:,space:]'
5	Display suspicious traffic excluding connections to 192.168.x.x	grep -Ei 'suspicious attack malicious intrusion' security_log.txt grep -Ev '192\.168\.[0-9]{1,3}\.[0-9]{1,3}'
6	Extract email addresses except those belonging to the organization's domain	grep -Eo '[:alnum:]._%+-]+@[[:alnum:].-]+\.[[:alpha:]]+' security_log.txt grep -Ev '@organization\.com\$'

Part G – Redirection

No.	Task	Command
1	Extract all IPv4 addresses → ips.txt	grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt > ips.txt
2	Extract all email addresses → emails.txt	grep -Eo '[[[:alnum:]]_%-]+@[[:alnum:]]-+\. [[[:alpha:]]]+' security_log.txt > emails.txt
3	Extract all URLs → urls.txt	grep -Eo 'https?://[^[:space:]]+' security_log.txt > urls.txt
4	Extract all hashes → hashes.txt	grep -Eoi '\b[a-f0-9]{32,64}\b' security_log.txt > hashes.txt
5	Extract failed-login entries → failed_login.txt	`grep -Ei 'failed.*login
6	Extract suspicious entries and append → incident_report.txt	grep -Ei 'suspicious attack malware intrusion exploit' security_log.txt >> incident_report.txt
7	Store unique IP addresses → unique_ips.txt	grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt sort -u > unique_ips.txt
8	Store IP-frequency statistics → ip_frequency.txt	grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt sort uniq -c > ip_frequency.txt
9	Normal output → results.txt, errors → errors.txt	grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt > results.txt 2> errors.txt
10	Domains from Domain accessed, lowercase, sorted, unique → domains.txt	grep -i 'Domain accessed' security_log.txt grep -Eo '([[[:alnum:]]-+\.)+ [[[:alpha:]]]+' tr '[:upper:]' '[:lower:]' sort -u > domains.txt
11	Unique IPs involved in failed logins → failed_login_ips.txt	grep -Ei 'failed.*login login.*failed authentication.*failed' security_log.txt grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' sort -u > failed_login_ips.txt

Combining Filters

No.	Task	Command
12	Extract failed-login usernames → failed_users.txt	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo 'User(name)?[=:] [[[:alnum:]]_-.]+' sed -E 's/^[^=:] +=:]// > failed_users.txt
13	Extract all domains, lowercase → domains.txt	grep -Eo '([[[:alnum:]]-+\.)+ [[[:alpha:]]]+' security_log.txt tr '[:upper:]' '[:lower:]' > domains.txt
14	Extract MD5 hashes, uppercase → md5_hashes.txt	grep -Eoi '\b[a-f0-9]{32}\b' security_log.txt tr '[:lower:]' '[:upper:]' > md5_hashes.txt
15	Failed-login source IPs, sorted and unique → failed_ips.txt	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' sort -u > failed_ips.txt
16	Remove all square brackets → clean_log.txt	tr -d '[' < security_log.txt > clean_log.txt
17	Convert complete log to lowercase → normalized_log.txt	tr '[:upper:]' '[:lower:]' < security_log.txt > normalized_log.txt
18	All domains → lowercase, sorted, unique → unique_domains.txt	grep -Eo '([[[:alnum:]]-+\.)+ [[[:alpha:]]]+' security_log.txt tr '[:upper:]' '[:lower:]' sort -u > unique_domains.txt
19	Failed-login entries → only usernames	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo 'User(name)?[=:] [[[:alnum:]]_-.]+' sed -E 's/^[^=:] +=:]//'

20	Failed-login entries → only source IPs, sorted	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' sort
21	IPs associated with failed logins → unique IPs	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' sort -u
22	Failed-login usernames → uppercase, unique	grep -Ei 'failed.*login login.*failed' security_log.txt grep -Eo 'User(name)?[=:]([[:alnum:]]_[-]+)' sed -E 's/^[^=:]++/[=:]/' tr '[:lower:]' '[:upper:]' sort -u
23	Domain accessed → domain names → lowercase	grep -i 'Domain accessed' security_log.txt grep -Eo '([[:alnum:]]-[\.])+[[:alpha:]]+' tr '[:upper:]' '[:lower:]'
24	MD5= → MD5 values → uppercase	grep -i 'MD5=' security_log.txt grep -Eoi 'MD5=[a-f0-9]{32}' cut -d= -f2 tr '[:lower:]' '[:upper:]'
25	User-Agent: → user-agent field → lowercase	grep -i 'User-Agent:' security_log.txt sed -E 's/.*User-Agent:[[:space:]]*/' tr '[:upper:]' '[:lower:]'
26	CVE-related lines → CVE identifiers → lowercase	grep -Eoi 'CVE-[0-9]{4}-[0-9]{4}' security_log.txt tr '[:upper:]' '[:lower:]'
27	Entries containing .exe → filename → uppercase	grep -Eoi '([[:alnum:]]_[-]+\.exe)' security_log.txt tr '[:lower:]' '[:upper:]'
28	Domain accessed → domains → uppercase, sorted, unique	grep -i 'Domain accessed' security_log.txt grep -Eo '([[:alnum:]]-[\.])+[[:alpha:]]+' tr '[:lower:]' '[:upper:]' sort -u